

MATH 8090:

Stationary Processes: Properties, Mean, and Covariance Functions

References: CC08 Chapter 2 & Chapter 4.1-4.3; BD16 Chapter
1.4, 1.5, 2.1, 2.2; SS17 Chapter 1.2-1.4

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Week 3

Agenda

Review

Mean and Autocovariance Functions

Stationarity

Some Examples of Stationary Processes

Linear Processes

Time Series Modeling Strategy

Additive Decompstion:

$$Y_t = \mu_t + s_t + \eta_t, \quad t = 1, 2, \dots, T$$

1. Plot the data y_t to explore the form of μ_t and s_t , and check for non-constant variation in η_t

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6. Check for fit of model (poor fit $\Rightarrow$ return to step 1)
7. Use model for inference: predicting future y_t 's, describing changes in y_t over time, hypothesis testing, etc

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- ▶ **Time series models** concern the modeling of temporal dependence in $\{\eta_t\}$

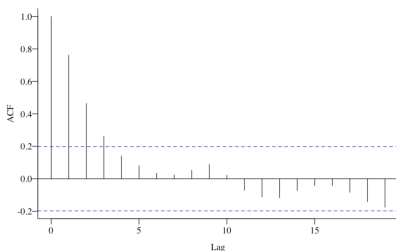
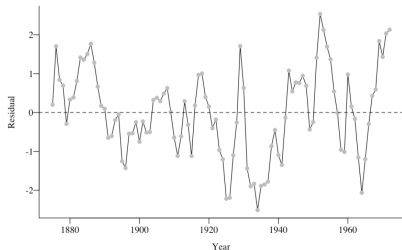
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- ▶ Residuals typically suggest **temporal dependence** in $\{\eta_t\}$
- ▶ **Time series models** concern the modeling of temporal dependence in $\{\eta_t\}$
- ▶ **Stationarity** assumption typically employed to overcome the issue of “one sample”
- ▶ **Weakly stationary**: constant mean and variance over time, with covariance depending only on time lags

The Implications of Temporal Dependence



- ▶ There is a consistent relationship between consecutive residuals
- ▶ The usual regression assumptions are violated, and t - and F -tests are not valid 😞
- ▶ We can get better predictions of future values by modeling autocorrelation 😊

Time Series Models

- ▶ A **time series model** is a specification of the probabilistic distribution of a sequence of random variables (RVs) $\{\eta_t\}_{t=1}^T$

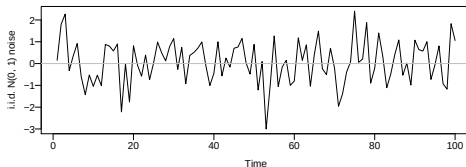
(The observed time series is a **realization** of such a sequence of random variables)

- ▶ The simplest time series is **i.i.d.** (*independent and identically distributed*) **noise**
 - ▶ $\{\eta_t\}$ is a sequence of independent and identically distributed zero-mean (i.e., $\mathbb{E}(\eta_t) = 0, \forall t$) random variables $\Rightarrow$ **no temporal dependence**
 - ▶ It is of little value of using i.i.d. noise model to conduct **forecast** as there is no information from the past observations
 - ▶ **But**, we will use i.i.d. model as a building block to develop time series models that can accommodate time dependence

Example Realizations of i.i.d. Noise

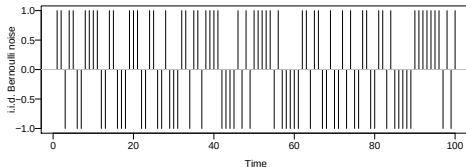
- Gaussian (normal) i.i.d. noise with mean 0 and variance $\sigma^2 > 0$

$$f(\eta_t) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\eta_t^2}{2\sigma^2}\right)$$



- Bernoulli i.i.d. noise with “success” probability

$$\mathbb{P}(\eta_t = 1) = p = 1 - \mathbb{P}(\eta_t = -1)$$



Means and Autocovariances

A time series may be characterized by its **mean function** and **autocovariance function**, which describe its first- and second-order structure

- ▶ The **mean function** of $\{\eta_t\}$ is

$$m_t = \mathbb{E}(\eta_t).$$

- ▶ m_t is the population mean at time t , which can be computed as:

$$m_t = \begin{cases} \int_{-\infty}^{\infty} \eta_t f(\eta_t) d\eta_t & \text{when } \eta_t \text{ is a continuous RV;} \\ \sum_{-\infty}^{\infty} \eta_t p(\eta_t), & \text{when } \eta_t \text{ is a discrete RV,} \end{cases}$$

where $f(\cdot)$ and $p(\cdot)$ are the probability density function and probability mass function of η_t , respectively

Examples of Mean Functions

- ▶ **Example 1:** What is the mean function for $\{\eta_t\}$, an i.i.d. $N(0, \sigma^2)$ process?
- ▶ **Example 2:** For each time point, let $Y_t = \beta_0 + \beta_1 t + \eta_t$ with β_0 and β_1 some constants and η_t is defined above. What is $m_Y(t)$?

Review: The Covariance Between Two RVs

- ▶ The **covariance** between the RVs X and Y is

$$\begin{aligned}\text{Cov}(X, Y) &= \mathbb{E}\{(X - \mu_X)(Y - \mu_Y)\} \\ &= \mathbb{E}(XY) - \mu_X\mu_Y.\end{aligned}$$

It is a measure of **linear dependence** between the two RVs.

When $X = Y$ we have

$$\text{Cov}(X, X) = \text{Var}(X).$$

- ▶ For constants a, b, c , and RVs X, Y, Z :

$$\begin{aligned}\text{Cov}(aX + bY + c, Z) &= \text{Cov}(aX, Z) + \text{Cov}(bY, Z) \\ &= a\text{Cov}(X, Z) + b\text{Cov}(Y, Z)\end{aligned}$$

$\Rightarrow$

$$\begin{aligned}\text{Var}(X + Y) &= \text{Cov}(X, X) + \text{Cov}(X, Y) + \text{Cov}(Y, X) + \text{Cov}(Y, Y) \\ &= \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)\end{aligned}$$

Autocovariance Function

- ▶ The **autocovariance function** of $\{\eta_t\}$ is

$$\gamma(s, t) = \text{Cov}(\eta_s, \eta_t) = \mathbb{E}[(\eta_s - m_s)(\eta_t - m_t)]$$

It measures the strength of **linear dependence** between two RVs η_s and η_t

- ▶ **Properties:**

- ▶ $\gamma(s, t) = \gamma(t, s)$ for each s and t
- ▶ When $s = t$ we have

$$\gamma(t, t) = \text{Cov}(\eta_t, \eta_t) = \text{Cov}(\eta_t) = \sigma_t^2$$

the value of the **variance function** at time t

- ▶ $\gamma(s, t)$ is a **non-negative definite** function (will come back to this later)

Autocorrelation Function

- ▶ The autocorrelation function of $\{\eta_t\}$ is

$$\rho(s, t) = \text{Corr}(\eta_s, \eta_t) = \frac{\gamma(s, t)}{\sqrt{\gamma(s, s)\gamma(t, t)}}$$

It measures the “scale invariant” linear association between η_s and η_t

- ▶ **Properties:**

- ▶ $-1 \leq \rho(s, t) \leq 1$ for each s and t
- ▶ $\rho(s, t) = \rho(t, s)$ for each s and t
- ▶ $\rho(t, t) = 1$ for each t
- ▶ $\rho(\cdot, \cdot)$ is a non-negative definite function

Why Stationarity Matters in Time Series

- ▶ We typically need “replicates” to estimate population quantities. For example, we use

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

as an estimate of μ_X , the population mean of the **single** RV, X

- ▶ However, in time series analysis, we have $n = 1$ (i.e., no replication), since we only observe one realized value at each time point
- ▶ **Stationarity** means that some characteristic of $\{\eta_t\}$ does not depend on the time point t , but only on the time lag between time points, **so that we can create “replicates.”**

Next, we will talk about **strict stationarity** and **weak stationarity**

Strictly Stationary Processes

- ▶ A time series, $\{\eta_t\}$, is **strictly stationary** if

$$[\eta_1, \eta_2, \dots, \eta_T] \stackrel{d}{=} [\eta_{1+h}, \eta_{2+h}, \dots, \eta_{T+h}],$$

for all integers h and $T \geq 1 \Rightarrow$ the **joint distribution** are unaffected by time shifts

- ▶ Under such the strict stationarity
 - ▶ $\{\eta_t\}$ is **identically distributed** but not (necessarily) **independent**
 - ▶ When μ_t is finite, $\mu_t = \mu$ is independent of time t
 - ▶ When the variance function exists,

$$\gamma(s, t) = \gamma(s + h, t + h),$$

for any s , t , and h

Weakly Stationary Processes

- ▶ $\{\eta_t\}$ is weakly stationary if
 - ▶ $\mathbb{E}(\eta_t) = \mu_t = \mu$
 - ▶ $\text{Cov}(\eta_t, \eta_{t+h}) = \gamma(t, t+h) = \gamma(h)$, finite constant that can depend on h but not on t
- ▶ Other names for this type of stationarity include second-order, covariance, wide sense. The quantity h is called the lag
- ▶ Weak and strict stationarity
 - ▶ A strictly stationary process $\{\eta_t\}$ is also weakly stationary as long as μ is finite
 - ▶ Weak stationarity does not imply strict stationarity!

Autocovariance Function of Stationary Processes

The autocovariance function (ACVF) of a stationary process $\{\eta_t\}$ is defined to be

$$\begin{aligned}\gamma(h) &= \text{Cov}(\eta_t, \eta_{t+h}) \\ &= \mathbb{E}[(\eta_t - \mu)(\eta_{t+h} - \mu)],\end{aligned}$$

which measures the lag- h time dependence

Properties of the ACVF:

- ▶ $\gamma(0) = \text{Var}(\eta_t)$
- ▶ $\gamma(-h) = \gamma(h)$ for each h
- ▶ $\gamma(s - t)$ as a function of $(s - t)$ is non-negative definite

Autocorrelation Function of Stationary Processes

The autocorrelation function (ACF) of a stationary process $\{\eta_t\}$ is defined to be

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)}$$

which measures the “scale invariant” lag- h time dependence

Properties of the ACF:

- ▶ $-1 \leq \rho(h) \leq 1$ and $\rho(0) = 1$ for each h
- ▶ $\rho(-h) = \rho(h)$ for each h
- ▶ $\rho(s - t)$ as a function of $(s - t)$ is non-negative definite

The White Noise Process $\{Z_t\}$

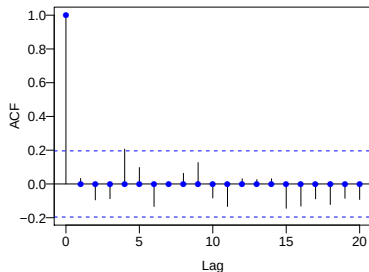
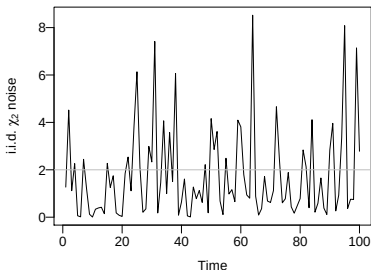
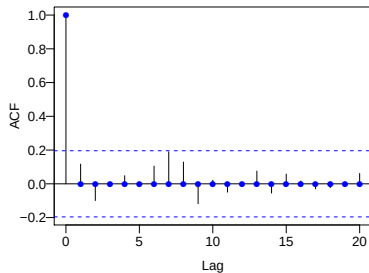
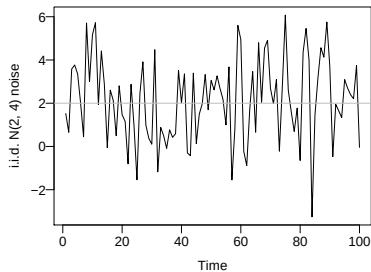
Let's assume $\mathbb{E}(Z_t) = \mu$ and $\text{Var}(Z_t) = \sigma^2 < \infty$. $\{Z_t\}$ is a **white noise** or **WN**(μ, σ^2) process if

$$\gamma(h) = 0,$$

for $h \neq 0$

- ▶ $\{Z_t\}$ is stationary
- ▶ However, distributions of Z_t and Z_{t+1} **can be different!**
- ▶ All i.i.d. noise with finite variance ($\sigma^2 < \infty$) is **white noise** **but the converse need not be true**

Examples Realizations of White Noise Processes



The Moving Average Process of First Order (MA(1))

Let $\{Z_t\}$ be a $WN(0, \sigma^2)$ process and θ be some constant $\in \mathbb{R}$.
For each integer t , let

$$\eta_t = Z_t + \theta Z_{t-1}.$$

- ▶ The sequences of RVs $\{\eta_t\}$ is called the **moving average process of order 1** or MA(1) process
- ▶ One can show that the MA(1) process $\{\eta_t\}$ is **stationary**

First-Order Moving Average Process: Mean Function

Need to show the mean function is NOT a function of time t

$$\begin{aligned}\mathbb{E}[\eta_t] &= \mathbb{E}[Z_t + \theta Z_{t-1}] \\ &= \mathbb{E}[Z_t] + \theta \mathbb{E}[Z_{t-1}] \\ &= 0 + \theta \times 0 \\ &= 0, \quad \forall t\end{aligned}$$



First-Order Moving Average Process: Covariance Function

Need to show the autocovariance function $\gamma(\cdot, \cdot)$ is a function of time lag only

$$\begin{aligned}\gamma(t, t+h) &= \text{Cov}(\eta_t, \eta_{t+h}) \\ &= \text{Cov}(Z_t + \theta Z_{t-1}, Z_{t+h} + \theta Z_{t+h-1}) \\ &= \text{Cov}(Z_t, Z_{t+h}) + \text{Cov}(Z_t, \theta Z_{t+h-1}) \\ &\quad + \text{Cov}(\theta Z_{t-1}, Z_{t+h}) + \text{Cov}(\theta Z_{t-1}, \theta Z_{t+h-1})\end{aligned}$$

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if $h = 0$, we have $\gamma(t, t+h) = \sigma^2 + \theta^2\sigma^2 = \sigma^2(1 + \theta^2)$

if $h = \pm 1$, we have $\gamma(t, t+h) = \theta\sigma^2$

if $|h| \geq 2$, we have $\gamma(t, t+h) = 0$

$\Rightarrow \gamma(t, t+h)$ only depends on h but not on t 😊

First-Order Moving Average Process: ACVF & ACF

ACVF:

$$\gamma(h) = \begin{cases} \sigma^2(1 + \theta^2) & h = 0; \\ \theta\sigma^2 & |h| = 1; \\ 0 & |h| \geq 2 \end{cases}$$

We can get **ACF** by dividing everything by $\gamma(0) = \sigma^2(1 + \theta^2)$

$$\rho(h) = \begin{cases} 1 & h = 0; \\ \frac{\theta}{1+\theta^2} & |h| = 1; \\ 0 & |h| \geq 2. \end{cases}$$

Examples Realizations of MA(1) Processes

First-order Autoregressive Process, AR(1)

Let $\{Z_t\}$ be a $WN(0, \sigma^2)$ process, and $-1 < \phi < 1$ be a constant.
Let's assume $\{\eta_t\}$ is a **stationary process** with

$$\eta_t = \phi\eta_{t-1} + Z_t,$$

for each integer t , where η_s and Z_t are **uncorrelated** for each $s < t \Rightarrow$ future noise is uncorrelated with the current time point

We will see later there is only one unique solution to this equation.
Such a sequence $\{\eta_t\}$ of RVs is called an **AR(1) process**

Properties of the AR(1) process

Want to find the mean value μ under the weakly stationarity assumption

$$\begin{aligned}\mathbb{E}[\eta_t] &= \mathbb{E}[\phi\eta_{t-1} + Z_t] \\ \mu &= \phi\mathbb{E}[\eta_{t-1}] + \mathbb{E}[Z_t] \\ \mu &= \phi\mu + 0 \\ \Rightarrow \mu &= 0, \quad \forall t\end{aligned}$$

Want to find $\gamma(h)$ under the weakly stationarity assumption

$$\begin{aligned}\text{Cov}(\eta_t, \eta_{t-h}) &= \text{Cov}(\phi\eta_{t-1} + Z_t, \eta_{t-h}) \\ \gamma(-h) &= \phi\text{Cov}(\eta_{t-1}, \eta_{t-h}) + \text{Cov}(Z_t, \eta_{t-h}) \\ \gamma(h) &= \phi\gamma(h-1) + 0 \\ \Rightarrow \gamma(h) &= \phi\gamma(h-1) = \dots = \phi^{|h|}\gamma(0)\end{aligned}$$

Properties of the AR(1) process Cont'd

$$\text{Var}(\eta_t) = \text{Var}(\phi\eta_{t-1} + Z_t)$$

$$\gamma(0) = \phi^2\gamma(0) + \sigma^2$$

$$\Rightarrow (1 - \phi^2)\gamma(0) = \sigma^2$$

$$\Rightarrow \gamma(0) = \frac{\sigma^2}{1 - \phi^2}$$

Therefore, we have

$$\gamma(h) = \begin{cases} \frac{\sigma^2}{1-\phi^2} & h = 0; \\ \frac{\phi^{|h|}\sigma^2}{1-\phi^2} & |h| \geq 1, \end{cases}$$

and

$$\rho(h) = \begin{cases} 1 & h = 0; \\ \phi^{|h|} & |h| \geq 1. \end{cases}$$

Examples Realizations of AR(1) Processes

The Random Walk Process

Let $\{Z_t\}$ be a $\text{WN}(0, \sigma^2)$ process and for $t \geq 1$ define

$$\eta_t = Z_1 + Z_2 + \cdots + Z_t = \sum_{s=1}^t Z_s.$$

- ▶ The sequence of RVs $\{\eta_t\}$ is called a **random walk process**
- ▶ **Special case:** If we have $\{Z_t\}$ such that for each t

$$\mathbb{P}(Z_t = z) = \begin{cases} \frac{1}{2}, & z = 1; \\ \frac{1}{2}, & z = -1, \end{cases}$$

then $\{\eta_t\}$ is a **simple symmetric random walk**

- ▶ The random walk process is not stationary!

Example Realizations of Random Walk Processes

Gaussian Processes

$\{\eta_t\}$ is a **Gaussian process (GP)** if the joint distribution of any collection of the RVs has a multivariate normal (aka Gaussian) distribution

- ▶ The distribution of a GP is fully characterized by $\mu(\cdot)$, the mean function, and $\gamma(\cdot, \cdot)$, the autocovariance function. The joint probability density function of $\boldsymbol{\eta} = (\eta_1, \eta_2, \dots, \eta_T)^T$ is

$$f(\boldsymbol{\eta}) = \frac{1}{(2\pi)^{\frac{T}{2}} |\Sigma|^{\frac{1}{2}}} \exp \left(-\frac{1}{2} (\boldsymbol{\eta} - \boldsymbol{\mu})^T \Sigma^{-1} (\boldsymbol{\eta} - \boldsymbol{\mu}) \right),$$

where $\boldsymbol{\mu} = (\mu_1, \mu_2, \dots, \mu_T)^T$ and the (i, j) element of the covariance matrix Σ is $\gamma(i, j)$

- ▶ If a GP $\{\eta_t\}$ is **weakly stationary** then the process is also **strictly stationary**

Gaussian Process Realizations under a Matérn Covariance

As an example, consider the [Matérn autocovariance function](#)

$$\gamma(h) = \sigma^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu}|h|}{\rho} \right)^\nu K_\nu \left(\frac{\sqrt{2\nu}|h|}{\rho} \right),$$

where σ^2 controls the variance, ρ controls the dependence range, and ν controls smoothness.

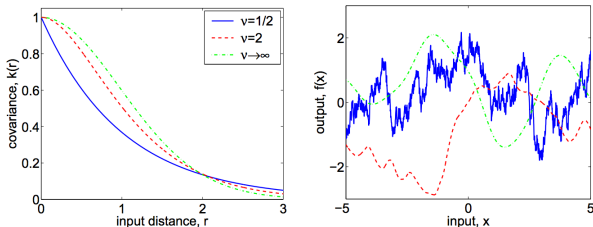


Figure: courtesy of Rasmussen & Williams 2006

Different realizations from the same GP share the same mean and autocovariance structure.

Linear Processes

- ▶ A time series $\{\eta_t\}$ is a **linear process** with mean μ if we can write it as

$$\eta_t = \mu + \sum_{j=-\infty}^{\infty} \psi_j Z_{t-j}, \quad \forall t,$$

where μ is a real-valued constant, $\{Z_t\}$ is a $\text{WN}(0, \sigma^2)$ process and $\{\psi_j\}$ is a set of absolutely summable constants¹

- ▶ Absolute summability of the constants guarantees that the infinite sum converges

¹A set of real-valued constants $\{\psi_j : j \in \mathbb{Z}\}$ is **absolutely summable** if $\sum_{j=-\infty}^{\infty} |\psi_j| < \infty$

Example: Moving Average Process of Order q , $MA(q)$

Let $\{Z_t\}$ be a $WN(0, \sigma^2)$ process. For an integer $q > 0$ and constants $\theta_1, \dots, \theta_q$ with $\theta_q \neq 0$, define

$$\begin{aligned}\eta_t &= Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q} \\ &= \theta_0 Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q} \\ &= \sum_{j=0}^q \theta_j Z_{t-j},\end{aligned}$$

where we let $\theta_0 = 1$

$\{\eta_t\}$ is known as the **moving average** process of order q , or the $MA(q)$ process, and, by definition, is a linear process

Defining Linear Processes with Backward Shifts

- ▶ Recall the backward shift operator, B , is defined by $B\eta_t = \eta_{t-1}$
- ▶ We can represent a linear process using the backward shift operator as $\eta_t = \mu + \psi(B)Z_t$, where we let $\psi(B) = \sum_{j=-\infty}^{\infty} \psi_j B^j$
- ▶ **Example:** we can write a mean zero MA(1) process as

$$\eta_t = \mu + \psi(B)Z_t,$$

where $\mu = 0$ and $\psi(B) = 1 + \theta B$

Linear Filtering Preserves Stationarity

- ▶ Let $\{Y_t\}$ be a time series and $\{\psi_j\}$ be a set of absolutely summable constants that does not depend on time
- ▶ **Definition:** A **linear time invariant** filtering of $\{Y_t\}$ with coefficients $\{\psi_j\}$ that do not depend on time is defined by

$$X_t = \psi(B)Y_t$$

- ▶ **Theorem:** Suppose $\{Y_t\}$ is a zero mean stationary series with ACVF $\gamma_Y(\cdot)$. Then $\{X_t\}$ is a zero mean stationary process with ACVF

$$\gamma_X(h) = \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \psi_j \psi_k \gamma_Y(j - k + h)$$

Example: The MA(q) Process is Stationary

By the filtering preserves stationarity result, the MA(q) process is a stationary process with mean zero and ACVF

$$\gamma(h) = \sigma^2 \sum_{j=0}^q \theta_j \theta_{j+h}$$

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By the filtering preserves stationarity result, the MA(q) process is a stationary process with mean zero and ACVF

$$\gamma(h) = \sigma^2 \sum_{j=0}^q \theta_j \theta_{j+h}$$

$$\begin{aligned} \gamma(h) &= \sum_{j=0}^q \sum_{k=0}^q \theta_j \theta_k \gamma_Z(j - k + h) \\ &= \sigma^2 \sum_{j=0}^q \sum_{k=0}^q \theta_j \theta_k \mathbb{1}(k = j + h) \\ &= \sigma^2 \sum_{j=0}^q \theta_j \theta_{j+h} \end{aligned}$$

Processes with a Correlation that Cuts Off

- ▶ A time series η_t is *q-correlated* if

η_t and η_s are uncorrelated $\forall |t - s| > q$,

i.e., $\text{Cov}(\eta_t, \eta_s) = 0, \forall |t - s| > q$

- ▶ A time series $\{\eta_t\}$ is *q-dependent* if

η_t and η_s are independent $\forall |t - s| > q$.

- ▶ **Theorem:** if $\{\eta_t\}$ is a stationary *q*-correlated time series with zero mean, then it can be always be represented as an $\text{MA}(q)$ process

The Autoregressive Process of Order p , $AR(p)$

- ▶ This process is attributed to [George Udny Yule](#). The $AR(1)$ process has also been called the Markov process
- ▶ Let $\{Z_t\}$ be a $WN(0, \sigma^2)$ process and let $\{\phi_1, \dots, \phi_p\}$ be a set of constants for some integer $p > 0$ with $\phi_p \neq 0$
- ▶ The $AR(p)$ process is defined to be the solution to the equation

$$\eta_t = \sum_{j=1}^p \phi_j \eta_{t-j} + Z_t \Rightarrow \underbrace{\eta_t - \sum_{j=1}^p \phi_j \eta_{t-j}}_{\phi(B)\eta_t} = Z_t,$$

where we let $\phi(B) = 1 - \sum_{j=1}^p \phi_j B^j$

A Stationary Solution for AR(1)

- ▶ We want the solution to the AR equation to yield a **stationary process**. Let's first consider AR(1). We will demonstrate that **a stationary solution exists for $|\phi_1| < 1$** .
- ▶ We first write

$$\begin{aligned}\eta_t &= \phi_1 \eta_{t-1} + Z_t = \phi_1(\phi_1 \eta_{t-2} + Z_{t-1}) + Z_t \\ &= \phi_1^2 \eta_{t-2} + \phi_1 Z_{t-1} + Z_t\end{aligned}$$

$\vdots$

$$= \phi_1^k \eta_{t-k} + \sum_{j=0}^{k-1} \phi_1^j Z_{t-j}$$

$\vdots$

$$= \sum_{j=0}^{\infty} \phi_1^j Z_{t-j}$$

AR(1) Example Cont'd

- Now let $\psi_j = \phi_1^j$. We then have

$$\eta_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}.$$

Using the fact that, for $|a| < 1$, $\sum_{j=0}^{\infty} a^j = \frac{1}{1-a}$, the sequence $\{\psi_j\}$ is absolutely summable

- Thus, since $\{\eta_t\}$ is a **linear process**, it follows by the filtering preserves stationarity result that $\{\eta_t\}$ is a zero mean stationary process with ACVF

$$\begin{aligned}\gamma(h) &= \sigma^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+h} \\ &= \sigma^2 \sum_{j=0}^{\infty} \phi_1^j \phi_1^{j+h} \\ &= \sigma^2 \phi_1^h \sum_{j=0}^{\infty} (\phi_1^2)^j\end{aligned}$$

AR(1) Example Cont'd

Now $|\phi_1| < 1$ implies that $|\phi_1^2| < 1$ and therefore we have

$$\gamma(h) = \frac{\sigma^2 \phi_1^h}{1 - \phi_1^2}$$

When $|\phi_1| \geq 1$

- ▶ No stationary solutions exist for $|\phi_1| = 1$
- ▶ When $|\phi_1| > 1$, dividing by ϕ_1 for both sides we get

$$\begin{aligned}\phi_1^{-1} \eta_t &= \eta_{t-1} + \phi_1^{-1} Z_t \\ \Rightarrow \eta_{t-1} &= \phi_1^{-1} \eta_t - \phi_1^{-1} Z_t\end{aligned}$$

A linear combination of **future** Z_t 's $\Rightarrow$ we have a stationary solution, **but**, η_t depends on future $\{Z_t\}$'s—This process is said to be not **causal**

- ▶ If we assume that η_s and Z_t are uncorrelated for each $t > s$, $|\phi_1| < 1$ is the only stationary solution to the AR equation

The Autoregressive Operator

- ▶ AR(1) process

$$\eta_t = \phi_1 \eta_{t-1} + Z_t \Rightarrow (1 - \phi_1 B) \eta_t = Z_t \Rightarrow \eta_t = (1 - \phi_1 B)^{-1} Z_t$$

- ▶ Recall $\sum_{j=0}^{\infty} a^j = \frac{1}{1-a} = (1-a)^{-1}$. We have

$$\eta_t = \sum_{j=0}^{\infty} (\phi_1 B)^j Z_t = \sum_{j=0}^{\infty} \phi_1^j B^j Z_t = \sum_{j=0}^{\infty} \phi_1^j Z_{t-j}$$

$\Rightarrow$ **This is another way to show that AR(1) is a linear process**

- ▶ Here $1 - \phi_1 B$ is the **AR characteristic polynomial**